

Version 2 block-download scheduler specification.

Models the layer above the per-connection protocol in *protocol.tla*: the node choosing which peer to ask for each block — the layer where Zebra’s genesis-to-tip sync stall lived in the legacy protocol (*ZcashFoundation/zebra*  $\neq$  10679; see *../sync\_scheduler.tla*). Zebra’s *v2* stack defers this layer to a future scheduler (“ibd-engine”), so this module models it ahead of the implementation.

The *v2* protocol gives a request more ways to end without a block than the legacy protocol did. Only one of them says anything about the peer’s chain:

|            |                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| not-found  | the per-entry result 0x02 of get-blocks: an explicit statement that the peer lacks the block.                                                                                     |
| REFUSED    | the responder reset the stream, declining to serve it (overload, rate limits, Zebra’s bulk-stream cap). Says nothing about the peer’s chain.                                      |
| truncation | the responder finished after any complete entry, leaving later entries unanswered (“get-blocks”, “get-block-range”). Routine and sanctioned; says nothing about the peer’s chain. |
| timeout    | no response within the stall timeout; the requester cancels with <i>CANCELLED</i> (“Block Download Parameters”). Says nothing about the peer’s chain.                             |

Three booleans mark each of the uninformative outcomes as informative, each reproducing a variant of the legacy registry-poisoning bug:

|                                |                             |                                                                            |
|--------------------------------|-----------------------------|----------------------------------------------------------------------------|
| <i>TreatRefusedAsMissing</i>   | $= \text{TRUE} \rightarrow$ | a REFUSED marks the peer missing                                           |
| <i>TreatTruncatedAsMissing</i> | $= \text{TRUE} \rightarrow$ | an unanswered entry marks the peer missing                                 |
| <i>BuggyTimeout</i>            | $= \text{TRUE} \rightarrow$ | a timeout marks the peer missing (the exact legacy zebra $\neq$ 10679 bug) |

Separately, the scheduler applies Zebra’s unresponsive-peer rule: a peer that times out *UnresponsiveLimit* requests in a row, with no response of any kind in between, is disconnected (*ZcashFoundation/zebra*  $\neq$  11276). The *Redial* boolean decides whether disconnected peers are ever redialled; without it, evicting the only holder of a block stalls the sync even though every registry entry is honest.

*RegistryHonest* and *EventuallyAllVerified* hold with every switch in its fixed position (*Redial* = TRUE, the rest FALSE) and are violated under the buggy settings, each as a short *TLC* counterexample.

See *../documents/v2-modeling.md* for the full write-up.

EXTENDS *Naturals*, *FiniteSets*

#### CONSTANTS

set of peer ids  
@type: *Set(Str)*;

*Peers*,

blocks to download are 1 .. *MaxBlock*  
@type: *Int*;

*MaxBlock*,

peers that hold only blocks 1 .. *LagTip*  
@type: *Set(Str)*;

*LaggingPeers*,

chain height of a lagging peer ( $< \text{MaxBlock}$ )  
@type: *Int*;

*LagTip*,  
 bound on re-queue attempts per block after a notfound  
 @type: *Int*;  
*MaxRetries*,  
 consecutive timeouts before a peer is disconnected  
 @type: *Int*;  
*UnresponsiveLimit*,  
 TRUE treats a REFUSED as a notfound (buggy)  
 @type: *Bool*;  
*TreatRefusedAsMissing*,  
 TRUE treats an unanswered entry as a notfound (buggy)  
 @type: *Bool*;  
*TreatTruncatedAsMissing*,  
 TRUE treats a timeout as a notfound (legacy bug)  
 @type: *Bool*;  
*BuggyTimeout*,  
 TRUE redials disconnected peers  
 @type: *Bool*;  
*Redial*

*Blocks*  $\triangleq 1 \dots \text{MaxBlock}$   
*NONE*  $\triangleq$  “none”    sentinel for “no peer”; must not be a member of *Peers*

Ground truth: a lagging peer holds blocks up to *LagTip*; every other peer is at tip and holds them all. The scheduler never reads this directly — it learns a peer’s contents only through the outcomes below.

*PeerTip*(*p*)  $\triangleq$  IF *p*  $\in$  *LaggingPeers* THEN *LagTip* ELSE *MaxBlock*  
*Holds*(*p*, *b*)  $\triangleq b \leq \text{PeerTip}(p)$

VARIABLES

@type: *Set(Int)*;  
*verified*,    subset of *Blocks* downloaded and verified (the frontier)  
 @type: *Set(Int)*;  
*pending*,    subset of *Blocks* the scheduler still wants and may request  
 @type: *Int*  $\rightarrow$  *Str*;  
*inflight*,    [ *Blocks*  $\rightarrow$  *Peers*  $\cup$  {*NONE*} ] peer a block is requested from  
 @type: *Str*  $\rightarrow$  (*Int*  $\rightarrow$  *Str*);  
*avail*,    [ *Peers*  $\rightarrow$  [ *Blocks*  $\rightarrow$  {“has”, “missing”, “unknown”} ] ] registry  
 @type: *Int*  $\rightarrow$  *Int*;  
*retries*,    [ *Blocks*  $\rightarrow 0 \dots \text{MaxRetries}$  ] re-queue attempts used  
 @type: *Str*  $\rightarrow$  *Int*;  
*timeouts*,    [ *Peers*  $\rightarrow 0 \dots \text{UnresponsiveLimit}$  ] consecutive unanswered timeouts  
 @type: *Str*  $\rightarrow$  *Bool*;  
*connected*    [ *Peers*  $\rightarrow$  BOOLEAN ] FALSE once disconnected as unresponsive

*vars*  $\triangleq \langle \text{verified}, \text{pending}, \text{inflight}, \text{avail}, \text{retries}, \text{timeouts}, \text{connected} \rangle$

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*Init*  $\triangleq$

$\wedge \text{verified} = \{\}$   
 $\wedge \text{pending} = \text{Blocks}$   
 $\wedge \text{inflight} = [b \in \text{Blocks} \mapsto \text{NONE}]$   
 $\wedge \text{avail} = [p \in \text{Peers} \mapsto [b \in \text{Blocks} \mapsto \text{"unknown"}]]$   
 $\wedge \text{retries} = [b \in \text{Blocks} \mapsto 0]$   
 $\wedge \text{timeouts} = [p \in \text{Peers} \mapsto 0]$   
 $\wedge \text{connected} = [p \in \text{Peers} \mapsto \text{TRUE}]$

The scheduler opens a get-blocks request for a pending block toward a connected peer not already marked missing for it.

*Request*  $\triangleq$

$\exists b \in \text{pending} :$   
 $\exists p \in \text{Peers} :$   
 $\wedge \text{inflight}[b] = \text{NONE}$   
 $\wedge \text{connected}[p]$   
 $\wedge \text{avail}[p][b] \neq \text{"missing"}$   
 $\wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = p]$   
 $\wedge \text{pending}' = \text{pending} \setminus \{b\}$   
 $\wedge \text{UNCHANGED } \langle \text{verified}, \text{avail}, \text{retries}, \text{timeouts}, \text{connected} \rangle$

The in-flight peer genuinely has the block: it is delivered and verified.

Any response resets the peer's consecutive-timeout count.

*DeliverBlock*  $\triangleq$

$\exists b \in \text{Blocks} :$   
 $\text{LET } p \triangleq \text{inflight}[b] \text{ IN}$   
 $\wedge p \neq \text{NONE}$   
 $\wedge \text{Holds}(p, b)$   
 $\wedge \text{verified}' = \text{verified} \cup \{b\}$   
 $\wedge \text{avail}' = [\text{avail} \text{ EXCEPT } ![p][b] = \text{"has"}]$   
 $\wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = \text{NONE}]$   
 $\wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = 0]$   
 $\wedge \text{UNCHANGED } \langle \text{pending}, \text{retries}, \text{connected} \rangle$

The in-flight peer genuinely lacks the block and answers the entry with the explicit not-found result. This is the one outcome that legitimately marks the registry, and the block is re-queued (bounded) toward other peers.

*NotFound*  $\triangleq$

$\exists b \in \text{Blocks} :$   
 $\text{LET } p \triangleq \text{inflight}[b] \text{ IN}$   
 $\wedge p \neq \text{NONE}$   
 $\wedge \neg \text{Holds}(p, b)$   
 $\wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = \text{NONE}]$   
 $\wedge \text{avail}' = [\text{avail} \text{ EXCEPT } ![p][b] = \text{"missing"}]$

$\wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = 0]$   
 $\wedge \text{IF } \text{retries}[b] < \text{MaxRetries}$   
     THEN  $\wedge \text{pending}' = \text{pending} \cup \{b\}$   
          $\wedge \text{retries}' = [\text{retries} \text{ EXCEPT } ![b] = @ + 1]$   
     ELSE  $\wedge \text{UNCHANGED } \text{pending}$  retries exhausted  
          $\wedge \text{UNCHANGED } \text{retries}$   
 $\wedge \text{UNCHANGED } \langle \text{verified}, \text{connected} \rangle$

The responder resets the request stream with REFUSED — it declines to serve the request, whether or not it holds the block (draft: “Request Streams”;  
Zebra refuses beyond 2 concurrent bulk streams). A REFUSED is a response,  
so the timeout count resets; what it does to the registry is the  
*TreatRefusedAsMissing* switch.

$\text{Refused} \triangleq$   
 $\exists b \in \text{Blocks} :$   
     LET  $p \triangleq \text{inflight}[b]$  IN  
      $\wedge p \neq \text{NONE}$   
      $\wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = \text{NONE}]$   
      $\wedge \text{pending}' = \text{pending} \cup \{b\}$   
      $\wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = 0]$   
      $\wedge \text{avail}' = [\text{avail} \text{ EXCEPT } ![p][b] =$   
         IF *TreatRefusedAsMissing* THEN “missing” ELSE @]  
      $\wedge \text{UNCHANGED } \langle \text{verified}, \text{retries}, \text{connected} \rangle$

The responder finishes the stream early, leaving this entry unanswered  
(draft: “get-blocks” / “get-block-range” allow finishing after any complete  
entry; truncation is resumable). Sanctioned and routine — but uninformative  
about the peer’s chain. The registry effect is the *TreatTruncatedAsMissing*  
switch.

$\text{Truncated} \triangleq$   
 $\exists b \in \text{Blocks} :$   
     LET  $p \triangleq \text{inflight}[b]$  IN  
      $\wedge p \neq \text{NONE}$   
      $\wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = \text{NONE}]$   
      $\wedge \text{pending}' = \text{pending} \cup \{b\}$   
      $\wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = 0]$   
      $\wedge \text{avail}' = [\text{avail} \text{ EXCEPT } ![p][b] =$   
         IF *TreatTruncatedAsMissing* THEN “missing” ELSE @]  
      $\wedge \text{UNCHANGED } \langle \text{verified}, \text{retries}, \text{connected} \rangle$

No response within the stall timeout: the requester cancels the stream with  
*CANCELLED* and re-queues the block (draft: “Block Download Parameters”).  
The registry effect is the *BuggyTimeout* switch — the exact legacy bug.  
The peer’s consecutive-timeout count rises; at *UnresponsiveLimit* the peer  
is disconnected as unresponsive (Zebra’s rule).

$\text{Timeout} \triangleq$

$$\begin{aligned}
& \exists b \in \text{Blocks} : \\
& \quad \text{LET } p \triangleq \text{inflight}[b] \text{ IN} \\
& \quad \wedge p \neq \text{NONE} \\
& \quad \wedge \text{inflight}' = [\text{inflight} \text{ EXCEPT } ![b] = \text{NONE}] \\
& \quad \wedge \text{pending}' = \text{pending} \cup \{b\} \\
& \quad \wedge \text{avail}' = [\text{avail} \text{ EXCEPT } ![p][b] = \\
& \quad \quad \text{IF } \text{BuggyTimeout} \text{ THEN "missing" ELSE } @] \\
& \quad \wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = \text{IF } @ < \text{UnresponsiveLimit} \text{ THEN } @ + 1 \text{ ELSE } @] \\
& \quad \wedge \text{connected}' = [\text{connected} \text{ EXCEPT } ![p] = @ \wedge \text{timeouts}[p] + 1 < \text{UnresponsiveLimit}] \\
& \quad \wedge \text{UNCHANGED } \langle \text{verified}, \text{retries} \rangle
\end{aligned}$$

A disconnected peer is redialled (Zebra maintains connections to its initial and discovered peers, redialling them when they fail). Gated by the *Redial* switch; without it eviction is forever.

$$\begin{aligned}
\text{Reconnect} & \triangleq \\
& \exists p \in \text{Peers} : \\
& \quad \wedge \text{Redial} \\
& \quad \wedge \neg \text{connected}[p] \\
& \quad \wedge \text{connected}' = [\text{connected} \text{ EXCEPT } ![p] = \text{TRUE}] \\
& \quad \wedge \text{timeouts}' = [\text{timeouts} \text{ EXCEPT } ![p] = 0] \\
& \quad \wedge \text{UNCHANGED } \langle \text{verified}, \text{pending}, \text{inflight}, \text{avail}, \text{retries} \rangle
\end{aligned}$$


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$$\begin{aligned}
\text{Next} & \triangleq \\
& \vee \text{Request} \\
& \vee \text{DeliverBlock} \\
& \vee \text{NotFound} \\
& \vee \text{Refused} \\
& \vee \text{Truncated} \\
& \vee \text{Timeout} \\
& \vee \text{Reconnect}
\end{aligned}$$

Strong fairness on the progress actions encodes the real-world assumption that the node keeps trying, a willing peer eventually responds, and the redial loop keeps running. Refused, *Truncated* and *Timeout* get no fairness — they are adversarial but never forced.

$$\begin{aligned}
\text{Spec} & \triangleq \\
& \text{Init} \\
& \wedge \Box [\text{Next}]_{\text{vars}} \\
& \wedge \text{SF}_{\text{vars}}(\text{Request}) \\
& \wedge \text{SF}_{\text{vars}}(\text{DeliverBlock}) \\
& \wedge \text{SF}_{\text{vars}}(\text{NotFound}) \\
& \wedge \text{SF}_{\text{vars}}(\text{Reconnect})
\end{aligned}$$


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Liveness: every block is eventually downloaded and verified.  
 $EventuallyAllVerified \triangleq \Diamond(\text{verified} = \text{Blocks})$

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Safety invariants.

$TypeOK \triangleq$

- $\wedge \text{verified} \subseteq \text{Blocks}$
- $\wedge \text{pending} \subseteq \text{Blocks}$
- $\wedge \text{inflight} \in [\text{Blocks} \rightarrow \text{Peers} \cup \{NONE\}]$
- $\wedge \text{avail} \in [\text{Peers} \rightarrow [\text{Blocks} \rightarrow \{\text{"has"}, \text{"missing"}, \text{"unknown"}\}]]$
- $\wedge \text{retries} \in [\text{Blocks} \rightarrow 0 \dots \text{MaxRetries}]$
- $\wedge \text{timeouts} \in [\text{Peers} \rightarrow 0 \dots \text{UnresponsiveLimit}]$
- $\wedge \text{connected} \in [\text{Peers} \rightarrow \text{BOOLEAN}]$

The registry never marks a peer missing for a block it actually holds:  
 only the explicit not-found result may mark it. “A refusal, a truncated  
 response, and a timeout are not notfound.”

$RegistryHonest \triangleq$

- $\forall p \in \text{Peers} :$
- $\forall b \in \text{Blocks} :$
- $\text{avail}[p][b] = \text{"missing"} \Rightarrow \neg \text{Holds}(p, b)$

A peer is only ever recorded as having a block it genuinely holds.

$RegistryHasIsSound \triangleq$

- $\forall p \in \text{Peers} :$
- $\forall b \in \text{Blocks} :$
- $\text{avail}[p][b] = \text{"has"} \Rightarrow \text{Holds}(p, b)$

A verified block was genuinely available from some peer.

$VerifiedAreReal \triangleq$

- $\forall b \in \text{verified} :$
- $\exists p \in \text{Peers} : \text{Holds}(p, b)$

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